

DBMS LAB - 3 : FUNCTIONS

```
CREATE TABLE STUDENT (  
    Student_ID INT PRIMARY KEY,  
    Student_Name VARCHAR(50),  
    Class VARCHAR(10),  
    Marks INT,  
    Fee_Paid DECIMAL(10,2),  
    Date_of_Birth DATE,  
    Admission_Date DATE  
);  
  
INSERT INTO STUDENT VALUES  
(101, 'Pratham', '10A', 85, 50000, '2008-05-15', '2023-06-10'),  
(102, 'Shreya', '10A', 92, 50000, '2008-11-20', '2023-06-12'),  
(103, 'Aditya Kumar', '10B', 76, 45000, '2009-02-10', '2023-06-11'),  
(104, 'Ananya Singh', '10B', 88, 48000, '2008-08-25', '2023-06-15'),  
(105, 'Rahul Verma', '10A', 65, 40000, '2009-01-30', '2023-06-13');  
  
SELECT * FROM STUDENT;  
  
SELECT Student_Name,  
    UPPER(Student_Name) AS UpperName  
FROM STUDENT;  
  
SELECT Student_Name,  
    LOWER(Student_Name) AS LowerNames  
FROM STUDENT;  
  
SELECT Student_Name,  
    LENGTH(Student_Name) AS Name_Length  
FROM STUDENT;  
  
SELECT CONCAT(Student_Name, Class) AS Student_Details  
FROM STUDENT;  
  
SELECT Student_Name,  
    SUBSTRING(Student_Name, 1, 5) AS First_Five  
FROM STUDENT;
```

```
SELECT TRIM(Student_Name) AS Trimmed_Name
FROM STUDENT;
```

```
SELECT Student_Name, Marks,
       ROUND(Marks * 1.05) AS Bonus
FROM STUDENT;
```

```
SELECT Student_Name, Marks,
       CEIL(Marks / 10.0) AS RoundUp
FROM STUDENT;
```

```
SELECT Student_Name, Marks,
       FLOOR(Marks / 10.0) AS Rounded_Down
FROM STUDENT;
```

```
SELECT Student_Name, Marks,
       MOD(Marks, 10) AS Remainder
FROM STUDENT;
```

```
SELECT Student_Name, Marks,
       ABS(Marks - 80) AS Difference_From_80
FROM STUDENT;
```

```
SELECT Student_Name, Marks,
       POWER(Marks, 2) AS Marks_Squared
FROM STUDENT;
```

```
SELECT Student_Name, Date_of_Birth,
       YEAR(Date_of_Birth) AS Birth_Year
FROM STUDENT;
```

```
SELECT Student_Name, Date_of_Birth,
       MONTH(Date_of_Birth) AS Birth_Month
FROM STUDENT;
```

```
SELECT Student_Name, Date_of_Birth,
```

```
        DAY(Date_of_Birth) AS Birth_Day
FROM STUDENT;

SELECT Student_Name, Date_of_Birth,
        TIMESTAMPDIFF(YEAR, Date_of_Birth, CURDATE()) AS Age
FROM STUDENT;

SELECT CURDATE() AS Today;


SELECT COUNT(*) AS Total_Students
FROM STUDENT;

SELECT SUM(Marks) AS Total_Marks
FROM STUDENT;

SELECT AVG(Marks) AS Average_Marks
FROM STUDENT;

SELECT MAX(Marks) AS Highest_Marks
FROM STUDENT;

SELECT MIN(Marks) AS Lowest_Marks
FROM STUDENT;

SELECT
    COUNT(*) AS Total_Students,
    SUM(Marks) AS Total_Marks,
    AVG(Marks) AS Average,
    MAX(Marks) AS Highest,
    MIN(Marks) AS Lowest
FROM STUDENT;
```

OUTPUT :

Output:

Student_ID	Student_Name	Class	Marks	Fee_Paid	Date_of_Birth	Admission_Date
101	Pratham	10A	85	50000.00	2008-05-15	2023-06-10
102	Shreya	10A	92	50000.00	2008-11-20	2023-06-12
103	Aditya Kumar	10B	76	45000.00	2009-02-10	2023-06-11
104	Ananya Singh	10B	88	48000.00	2008-08-25	2023-06-15
105	Rahul Verma	10A	65	40000.00	2009-01-30	2023-06-13

Student_Name	UpperName
Pratham	PRATHAM
Shreya	SHREYA
Aditya Kumar	ADITYA KUMAR
Ananya Singh	ANANYA SINGH
Rahul Verma	RAHUL VERMA

Student_Name	LowerName
Pratham	pratham
Shreya	shreya
Aditya Kumar	aditya kumar
Ananya Singh	ananya singh
Rahul Verma	rahul verma

Student_Name	Name_Length
Pratham	7
Shreya	6
Aditya Kumar	12
Ananya Singh	12
Rahul Verma	11

```

+-----+
| Student_Details |
+-----+
| Pratham10A      |
| Shreya10A       |
| Aditya Kumar10B |
| Ananya Singh10B |
| Rahul Verma10A  |
+-----+

```

```

+-----+-----+
| Student_Name | First_Five |
+-----+-----+
| Pratham      | Prath      |
| Shreya       | Shrey      |
| Aditya Kumar | Adity      |
| Ananya Singh | Anany      |
| Rahul Verma  | Rahul      |
+-----+-----+

```

```

+-----+
| Trimmed_Name |
+-----+
| Pratham      |
| Shreya       |
| Aditya Kumar |
| Ananya Singh |
| Rahul Verma  |
+-----+

```

```

+-----+-----+-----+
| Student_Name | Marks | Bonus |
+-----+-----+-----+
| Pratham      | 85    | 89    |
| Shreya       | 92    | 97    |
| Aditya Kumar | 76    | 80    |
| Ananya Singh | 88    | 92    |
| Rahul Verma  | 65    | 68    |
+-----+-----+-----+

```

Student_Name	Marks	RoundUp
Pratham	85	9
Shreya	92	10
Aditya Kumar	76	8
Ananya Singh	88	9
Rahul Verma	65	7

Student_Name	Marks	Rounded_Down
Pratham	85	8
Shreya	92	9
Aditya Kumar	76	7
Ananya Singh	88	8
Rahul Verma	65	6

Student_Name	Marks	Remainder
Pratham	85	5
Shreya	92	2
Aditya Kumar	76	6
Ananya Singh	88	8
Rahul Verma	65	5

Student_Name	Marks	Difference_From_80
Pratham	85	5
Shreya	92	12
Aditya Kumar	76	4
Ananya Singh	88	8
Rahul Verma	65	15

Student_Name	Marks	Marks_Squared
Pratham	85	7225
Shreya	92	8464
Aditya Kumar	76	5776
Ananya Singh	88	7744
Rahul Verma	65	4225

Student_Name	Date_of_Birth	Birth_Year
Pratham	2008-05-15	2008
Shreya	2008-11-20	2008
Aditya Kumar	2009-02-10	2009
Ananya Singh	2008-08-25	2008
Rahul Verma	2009-01-30	2009

Student_Name	Date_of_Birth	Birth_Month
Pratham	2008-05-15	5
Shreya	2008-11-20	11
Aditya Kumar	2009-02-10	2
Ananya Singh	2008-08-25	8
Rahul Verma	2009-01-30	1

Student_Name	Date_of_Birth	Birth_Day
Pratham	2008-05-15	15
Shreya	2008-11-20	20
Aditya Kumar	2009-02-10	10
Ananya Singh	2008-08-25	25
Rahul Verma	2009-01-30	30

Student_Name	Date_of_Birth	Age
Pratham	2008-05-15	18
Shreya	2008-11-20	17
Aditya Kumar	2009-02-10	17
Ananya Singh	2008-08-25	18
Rahul Verma	2009-01-30	17

Today
2026-08-25

Total_Students
5

Total_Marks
406

+-----+	
	Average_Marks
+-----+	
	81.2000
+-----+	

+-----+	
	Highest_Marks
+-----+	
	92
+-----+	

+-----+	
	Lowest_Marks
+-----+	
	65
+-----+	

+-----+		+-----+		+-----+		+-----+		+-----+	
	Total_Students	Total_Marks	Average	Highest	Lowest				
+-----+		+-----+		+-----+		+-----+		+-----+	
	5	406	81.2000	92	65				
+-----+		+-----+		+-----+		+-----+		+-----+	